

Developing spatial reasoning skills using virtuality puppets

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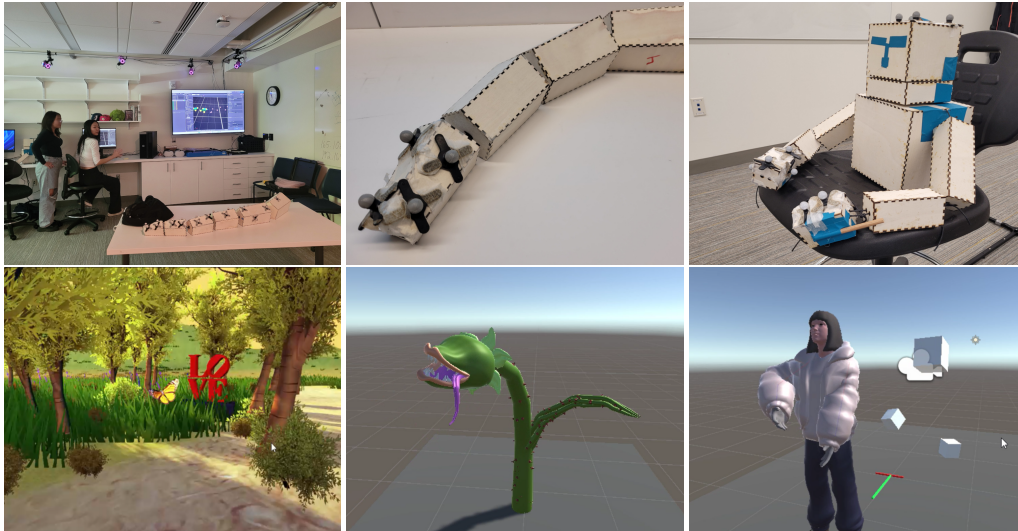


Figure 1: Top: Setting up the tentacle puppet for tracking; The tentacle and body puppets configured with optical markers. Bottom: The butterfly is controlled with a cardboard box; The head and tentacle are controlled using the tentacle puppet; The person's head and arms are controlled using the body puppet. Model credits: Gavin Sears, Neha Thumu, ReadyPlayerMe

Abstract

A virtuality puppet links a physical puppet with a virtual counterpart. In this work, we describe three projects that use virtuality puppets to help students develop skills for working with frames of reference and animating characters. In the first project, students setup a mapping between our motion capture space and the virtual environment. In the second and third, students track the motion of physical puppets, such as a tentacle or Muppet, and retarget the motion to different virtual characters. Each project reinforces lecture material about transformations, coordinate systems, and kinematics while also providing opportunities for hands on construction, 3D modeling, and scripting that allow students to show their creativity.

CCS Concepts

• Computing methodologies → Motion capture; Animation.

Keywords

Instructional materials, Motion tracking, Kinematics, Puppetry



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1 Overview

We created several hands-on activities to help students develop intuition for working with transformations and frames of reference. Our approach is based on the use of props and puppets tracked with an optical motion capture system and streamed into Unity. We call this approach virtuality puppetry because it links real-world objects – constructed with real-life materials – with virtual objects inside a game engine. Pedagogically, the virtuality setup allows students to understand the mathematical concepts needed to create the virtual world using real-world objects as a reference. Best of all, these projects offer ample opportunities for fun, teamwork, and creativity, for example, to create the props and puppets, to set-up and capture their movement, and to overlay imaginative characters and environments on top of them.

This work describes three projects that we prototyped as part of a virtual reality seminar at Bryn Mawr College (See Table 1 for our learning goals). The seminar included a mix of students with varying levels of graphics experience: some who had taken a computer

Table 1: Projects Summary

Project	Learning Goals
Project 1: Box	• Left-handed versus Right-handed systems • Mappings between coordinate frames • Transforms as sequences of scale, rotate, translate
Project 2: Tentacle	• Transformation hierarchies • Aligning coordinate systems • Interpolating rotations with slerp
Project 3: Body	• Aligning coordinate systems • Two-link analytic inverse kinematics

graphics course; some who had taken a Unity-based game development course, and some with modeling experience with Blender. These projects are intended to complement lecture and worksheet material about rotations in different representations (quaternions, matrix, and Euler angles), transformations, frames of reference, and forward and inverse kinematics. In the first project, students setup an accurate mapping between the motion capture space and the Unity scene. Students measured the space and created demos that could be loaded into VR. In the second and third projects, students worked with physical puppets: a tentacle and a Muppet. To start, students tracked the limbs of each puppet and mapped their motion directly to virtual limbs. To extend the controls, the students explored several retargeting strategies, such as orientation constraints, a gaze target, and two-link inverse kinematics. Students worked in groups of two, pairing people with complementary expertise. Each puppetry activity took approximately 1.5 hours to complete (not including puppet construction and 3D modeling), including 20 minutes to describe the activity and review the necessary math and algorithms.

2 Metadata

These materials are intended as a creative undergraduate project within a computer graphics, games, or animation course that is part of a computer science major. The ACM CCS classification terms are

- (1) Animation
- (3) Fundamentals

3 Materials

Puppet patterns and unity projects are available on Github¹.

Puppets We prototyped our puppets in cardboard and then transferred the designs for laser cutting. The cardboard puppets were constructed using standard delivery boxes and duck tape. To create the laser cut puppets we used box.py² to recreate the cardboard segments. The laser cut boxes are assembled from flat pieces of 3mm balsa wood that have square-tooth edges for assembly. We used wood glue to strengthen the assembly and joined the limbs using cords.

Motion tracking and streaming We used a Vicon optical motion capture system to track our props and puppets. The system can track a rigid body with four asymmetrically-placed optical markers.

¹<https://github.com/alinen/virtuality-puppets>

²<https://boxes.hackerspace-bamberg.de/>

Table 2: Metadata in tabular format.

Summary	Using real-world props and puppets to build intuition about theoretical concepts.
Learning Outcomes	Techniques for aligning frames of reference and tracking orientations and positions.
Classifications	Fundamentals: Transformations, Kinematic chains, Inverse Kinematics
Audience	Undergraduate CS education
Dependencies	CS2, Experience with a high-level language like Java or C#
Prerequisites	Matrices, vectors, homogeneous coordinates, transformations, and rotations.
Strengths	Students quickly became proficient in mapping the puppets to characters. They also enjoyed using the motion capture equipment, designing and tracking the puppets, and building their own virtual characters and environments.
Weaknesses	These activities require an optical motion capture system with real-time streaming features. Without real-time capture, students can still use the puppets to check measurements and visualize their coordinate systems. However, the ability to physically walk around and track the corresponding changes in Unity is very helpful. This activity must be done in small groups. Although we used a laser printer to construct our puppets, cardboard also works but wears out with a lot of handling.
Variants	Any number of puppets and virtual characters and environments are possible.
Assessments	Students are assessed based on completion, correct functionality, and originality

The origin and coordinate axes of the rigid body can be configured to align with the object to make the mappings to a virtual character easier.

Game Engine We used the Vicon Datastream SDK³ to stream the position and orientation of our props into Unity. The SDK automatically converts to the Unity coordinate system. However, it is still necessary to understand the mapping between the coordinate systems in order to accurately size the floor and puppets. The virtual characters were additionally designed and configured so that their limbs aligned with the orientations of their physical counterparts.

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³<https://www.vicon.com/software/datastream-sdk/>